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DRPM: An advanced predictive model for early diabetes detection and risk stratification

Fulei Nie,[1,2] Xiaoming Song,[2,4] and Wei Chen[1,3]

[1]School of Public Health, North China University of Science and Technology, Tangshan 063210, China; [2]School of Life Sciences, North China University of Science and Technology, Tangshan 063210, China; [3]Innovative Institute of Chinese Medicine and Pharmacy, Chengdu University of Traditional Chinese Medicine, Chengdu 611137, China; [4]School of Basic Medical Sciences, North China University of Science and Technology, Tangshan 063210, China

Diabetes is a prevalent chronic disease that poses a significant burden on individuals and healthcare systems. Early diagnosis and effective management are essential for delaying disease onset and minimizing complications. In this study, we developed a deep learning-based risk prediction model using data from the National Health and Nutrition Examination Survey (NHANES) spanning from 2011 to 2018. From an initial set of 36 variables, and through feature selection, five key features were identified through feature selection techniques to construct the model. The model demonstrated robust performance, accurately predicting diabetes risk with high precision in test data. External validation further confirmed its ability to correctly identify individuals at risk of developing diabetes. To enhance its practical application, we implemented a risk stratification system and developed a user-friendly online tool, available at http://cbcb.cdutcm.edu.cn/drpm/, allowing easy access for users. This model provides a valuable tool for diabetes risk screening and personalized early detection.

INTRODUCTION

Diabetes is a major global chronic disease that leads to severe complications and imposes a significant burden on individuals, families, and society. According to the International Diabetes Federation, approximately 530 million adults worldwide were affected by diabetes in 2021, with projections indicating an increase to 643 million by 2030 and 783 million by 2045.[1–3] Moreover, delayed diagnosis increases the risk of complications, further straining already overwhelmed healthcare systems.[4] Therefore, early prediction of diabetes onset is crucial for disease prevention and improved health outcomes.

Previous research has primarily focused on developing predictive models for diabetes. For instance, several studies[5–8] have examined genome-wide association marker-based risk models (GRMs) for assessing the risk of type 2 diabetes mellitus (T2DM). The advent of genome-wide association studies has significantly accelerated the identification of T2DM-associated loci, facilitating the development of genetic-based prediction models. Although these models have shown potential in identifying diabetes risk, GRMs have demonstrated relatively poor discrimination for T2DM.[9,10] There were also some studies that predict the risk of diabetes based on gut microbiota[11–13]; however, gut microbiota is highly susceptible to external influences such as diet, age, and medication, posing challenges for model generalizability and clinical application. With the rapid expansion of physical examination data and advancements in machine learning (ML) technologies, ML-based models using electronic health records (EHRs) have emerged as powerful tools for diabetes prediction.[14–16] However, most of these studies relied on conventional ML algorithms, such as support vector machines and random forest (RF).[17,18] For example, Zou et al.[19] developed an RF-based model to predict diabetes in the Chinese population, achieving an accuracy of 0.8084. However, their study did not introduce a systematic risk scorecard for diabetes assessment. In another study, Yang et al.[20] applied extreme gradient boosting (XGBoost) and incorporated key features such as BMI (body mass index), FBG (fasting blood glucose), age, WtHR (waist-to-height ratio), MSP (mean systolic pressure), and UGLU (urine glucose) for diabetes identification. Their model achieved an AUC (area under the receiver operating characteristic curve) of 0.8768, indicating strong predictive performance. However, the reliance on UGLU as a feature required hospital-based testing, limiting its practicality for personal use and early self-assessment.

In recent years, deep learning algorithms have been increasingly applied to EHRs for predicting disease onset.[21,22] Compared to traditional ML methods, deep learning models demonstrate superior performance in healthcare research due to their ability to handle the complex and high-dimensional nature of EHR data.[23–25] For diabetes prediction, deep learning methods efficiently extract valuable insights from EHRs by capturing relevant features, thereby enabling targeted interventions and facilitating lifestyle adjustments for at-risk individuals. Nguyen et al.[26] developed a deep learning model tailored for predicting diabetes in the US population, achieving an accuracy of 0.8428 and an AUC of 0.8413. However, its reliance on

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Correspondence: Wei Chen, School of Public Health, North China University of Science and Technology, Tangshan 063210, China.
E-mail: greatchen@ncst.edu.cn

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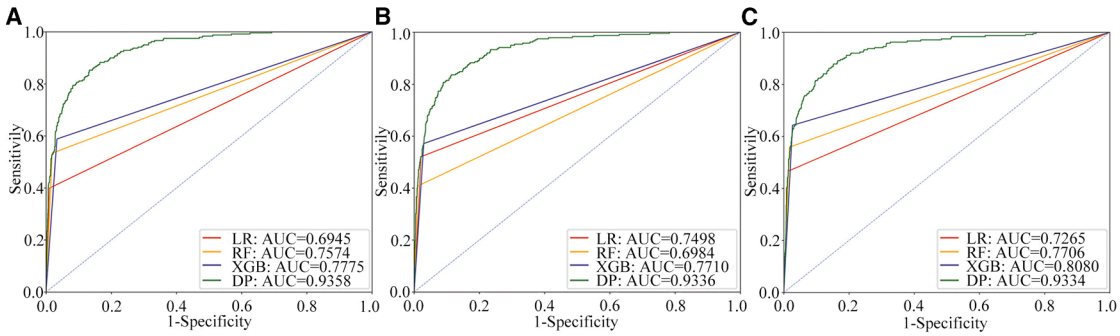


Figure 1. The ROC curves generated using different feature combinations

(A) ROC curves for the validation data based on five features, (B) ROC curves for the validation data based on seven features, and (C) ROC curves for the validation data based on thirty-six features.

a large number of features may limit its applicability in clinical settings. Zhang et al.[27] integrated fundus images with clinical metadata to develop deep learning models for predicting type 2 diabetes, achieving AUC values between 0.8280 and 0.9290. Despite the growing number of deep learning models for diabetes risk prediction using EHR data, few have provided clinically impactful tools for diabetes diagnosis, limiting their practical effectiveness in achieving their intended goals.

In the present work, we introduce a diabetes risk evaluation tool based on comprehensive physical examination data from National Health and Nutrition Examination Survey (NHANES), aiming to enhance healthcare accessibility by encouraging patient self-monitoring. The proposed diabetes risk prediction model (DRPM) relies on three key kinds of physical attributes; namely, demographics, vital signs, and blood and urine data. After feature selection, five features were selected to build the risk prediction model. The essential data required for this model can be easily collected at home with appropriate safety precautions, making it suitable for public health screening. To facilitate clinical use, a web-based tool was developed to compute the diabetes risk, which is freely accessible at http://cbcb.cdutcm.edu.cn/drpm/.

RESULTS

Selection of diabetes risk factors

To identify the smallest set of features that yields the most accurate predictions, two feature-ranking stages were employed. In the first stage, deep learning methods were used to evaluate the classification performance of each feature, followed by analysis of variance (ANOVA) to assess the significance of each feature. A single-input deep learning model was used as the primary classifier to evaluate the individual performance of each feature. The features were then ranked based on their AUC in descending order. In the second stage, the importance scores of all 36 features were calculated using ANOVA and sorted in descending order. Based on these significance scores, the top 10 features were selected (highlighted in the green box), while two inconsistent features (indicated by the blue underline) were removed (Figure S1). Among the selected features, WtHR was derived from waist circumference and BMI was derived from body weight. The features were processed by retaining the one with the higher AUC and discarding the one with the lower AUC. As a result, 7 features—FBG, age, WtHR, MSP, BMI, triglycerides, and albumin—were selected for model comparison.

Comparison of different classification algorithms

To determine the most effective algorithm for model development, we compared the performance of four common ML models: fully connected neural network (FCNN), XGBoost, logistic regression (LR), and RF. The receiver operating characteristic (ROC) curves for the validation data, based on different feature combinations, are shown in Figure 1. Regardless of whether the models were based on 5, 7, or 36 features, the AUCs for the LR, RF, and XGB models were consistently below 0.8 in the validation set, with the exception of the XGB model using 36 features, which achieved an AUC of 0.8080. The AUCs of the FCNN-based deep learning models in the validation set all exceeded 0.9. These results indicate that the FCNN-based deep learning model achieved the highest AUC across all feature combinations. Further details can be found in Table S1. Based on these findings, the FCNN method was selected to build the final deep learning model using five key features: FBG, age, WtHR, MSP, and BMI.

Performance evaluation in classifying healthy and diabetic individuals

The final deep learning model was constructed using the five optimal features identified through feature selection. The performance of the model is presented in Table 1 and Figure S2. Individuals with a predicted risk score (PRS) of 0.5 or lower were classified as “healthy,” while those with a PRS above 0.5 were classified as “diabetic.” The model achieved AUCs of 0.9399, 0.9358, and 0.9601 in the training, validation, and testing datasets, respectively. These results suggest that the model is highly effective at identifying individuals with diabetes and demonstrates robustness across different datasets. Other evaluation metrics are detailed in Table 1. As shown in the table, the specificity of the model across all datasets was high, indicating it can effectively identify the majority of non-diabetic individuals.

Table 1. Performances of the model for identifying diabetes patients

	AUC	Accuracy	Mcc	Sp	Sn	Precision	F1
Training	0.9399	0.9321	0.6277	0.9784	0.5701	0.7715	0.6557
Validation	0.9358	0.9282	0.5991	0.9791	0.5294	0.7636	0.6253
Testing	0.9601	0.9363	0.6580	0.9764	0.6218	0.7708	0.6884

However, the sensitivity was relatively lower. An analysis of the three datasets revealed that most diabetic individuals were taking medication, such as diabetic pills or insulin, to manage their blood sugar levels. A large proportion of these diabetic individuals, who had a PRS below 0.5, exhibited FBG levels below 7.0 mmol/L or HbA1c values under 6.5% (Table S2). This might be one of the factors contributing to the reduced sensitivity of the model.

To validate the efficacy of the five optimal features, we compared the final model with the results obtained using all available features (Table S3). The AUC of the final model was comparable to that of the model utilizing all features, demonstrating that the feature selection process did not compromise predictive performance. However, the model built with the five optimal features is more conducive to real-world applications. For example, this model enables users to assess their diabetes risk without needing hospital visits or laboratory tests, making it more accessible and practical for self-monitoring.

Diabetes risk stratification

To enhance the clinical applicability of the model, we introduced a risk stratification approach, which we designated as the DRPM. In line with the training process, a risk threshold of 0.5 was set to differentiate individuals at risk of diabetes. Specifically, individuals with a PRS above 0.5 were classified as high-risk. Individuals with a PRS of 0.13 or below were classified as low-risk, while those with a PRS between 0.13 and 0.5 were classified as moderate-risk. This classification threshold of 0.13 was determined based on the upper bound of the boxplot for healthy individuals in the training and validation datasets (Figure S3). This stratification facilitates personalized risk assessments, offering a more tailored approach to diabetes prediction.

The distribution of PRS and risk groups among participants in the three cohorts is illustrated in Figure 2 and Table 2, with Figures 2A–2C representing the training, validation, and testing datasets. In the training dataset (6,530 healthy records and 835 diabetes records), the proportion of individuals in the high-risk group was 8.38% (617 records), 11.65% (858 records) were in the moderate-risk group, and 79.97% (5,890 records) were in the low-risk group. Among the 617 individuals in the high-risk category, 476 were diagnosed with diabetes, accounting for 77.15% of all diabetes cases. Within the moderate-risk group, comprising 858 records, 230 (26.81% of all diabetes records) were diagnosed with diabetes. In the low-risk group, which comprised 5,890 records, 129 (2.19% of all diabetes records) were diagnosed with diabetes (Figures 2A–2C; Table 2). Notably, consistent patterns were observed across all three datasets, underscoring the robustness and reliability of our risk stratification approach. Individuals diagnosed with diabetes were predominantly classified in the high-risk group, whereas those without diabetes were mostly categorized in the low-risk category. This consistency in outcomes across the datasets further reinforces the effectiveness and validity of our risk stratification methodology.

External validation of the risk stratification model

The validation of the risk stratification model was conducted independently using the external validation cohort, consisting of 1,836 samples from the NHANES conducted between 2019 and March 2020. This cohort included 1,596 healthy individuals and 240 diabetic patients. The outcomes of risk stratification prediction using DRPM are presented in Figure 3.

Among the 1,836 analyzed instances, 9.37% (172 records) were classified as high-risk, 13.45% (243 records) as moderate-risk, and the majority, 77.18% (1,378 records), as low-risk. Within the high-risk group (172 records), 134 individuals (55.83% of all diabetic individuals) were identified as diabetic. Of note, 38 individuals in the high-risk group were non-diabetic, representing a false positive rate of 2.38% relative to the non-diabetic population. Upon further investigation of this high-risk non-diabetic group, 7 individuals were diagnosed with borderline diabetes, 16 with prediabetes, and 14 with no prior diagnosis of diabetes or prediabetes but exhibiting FBG or glycated hemoglobin levels suggestive of a prediabetic state. In total, 37 out of the 38 high-risk non-diabetic individuals showed tendencies toward developing diabetes, reinforcing the predictive capacity of our model for early-stage diabetes detection.

A similar pattern was observed in the moderate-risk group. Among the 247 individuals in this category, 67 were diabetic, and 180 were non-diabetic. Of the 180 non-diabetic individuals, 43.89% showed a tendency to develop diabetes, indicating that the model is effective at identifying individuals at risk of progression to diabetes. Finally, in the low-risk group (1,417 cases), 86.34% (1,378 individuals) were non-diabetic, only 39 individuals were misclassified, corresponding to a false negative rate of 16.25%. This highlights the model's capacity to accurately classify the majority of individuals with and without diabetes.

Overall, the external validation results confirm the robustness of the model and its ability to identify not only individuals with existing diabetes but also those at an early stage or at risk of developing diabetes.

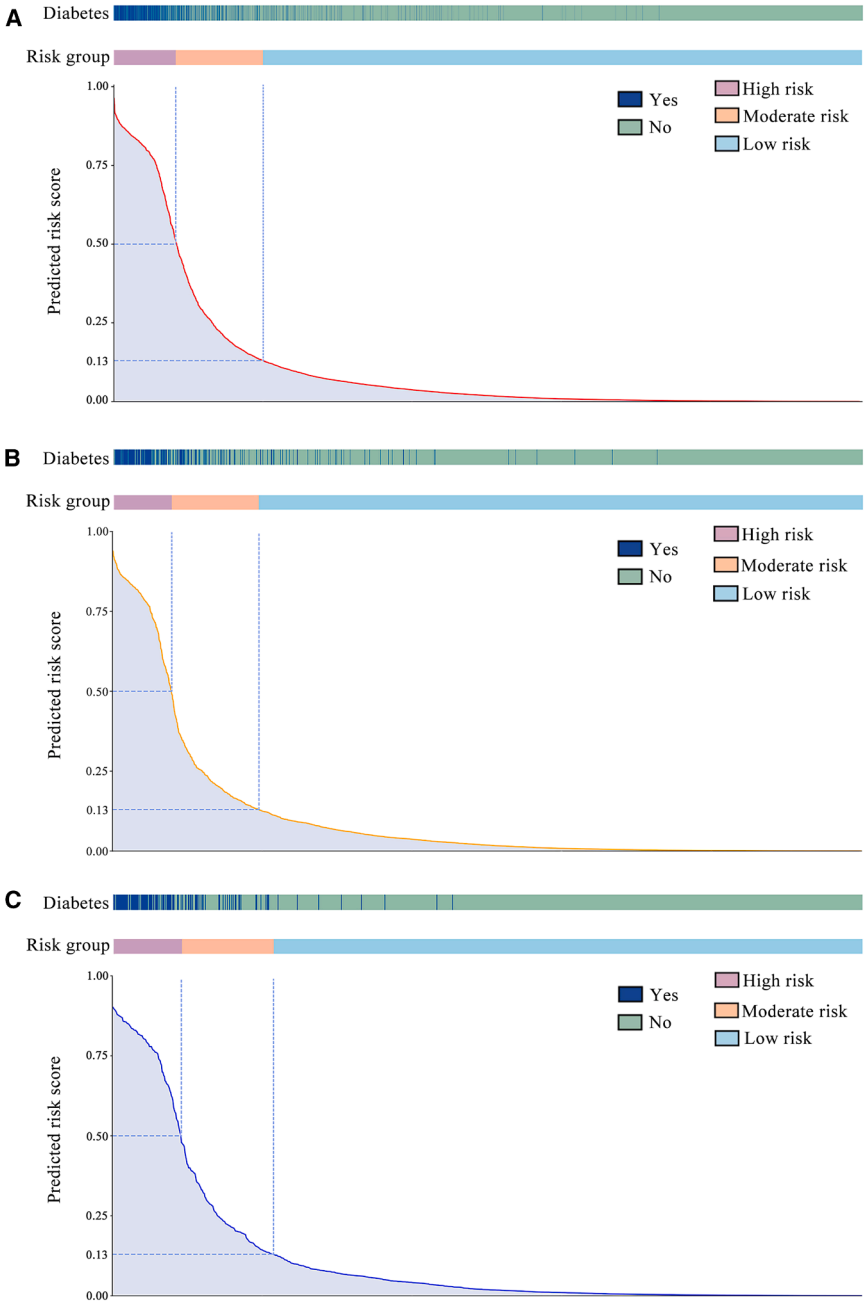


Figure 2. Evaluation of risk stratification using the final model

Distribution of predicted risk scores and risk groups in (A) training dataset, (B) validation dataset, and (C) testing dataset.

Model explainability

To address the "black box" nature of deep learning models, Shapley additive explanations (SHAP)[28] was incorporated to assess the contribution of each feature. The SHAP summary plot (Figure 4A) ranks features based on the mean absolute value of their global SHAP scores, in descending order of importance. The results revealed that FBG was the most significant contributing feature for both diabetic and healthy individuals. In the diabetic population, FBG accounted for 64% of the mean total SHAP value, while in the non-diabetic group, it contributed 48%. The top three features (FBG, age, and WtHR) together accounted for 89% of the total SHAP value in both groups.

As shown in Figure 4B, features are ranked along the *y* axis, with SHAP values plotted along the *x* axis to reflect their influence on the model's predictions. Each dot represents an individual sample, where the color indicates the magnitude of the original feature value—red for high and blue for low. Higher values of FBG, older age, and increased WtHR were all associated with elevated diabetes risk. The dot plots in Figures 4C and S4 illustrate the relationship between feature values and SHAP contributions. Among non-diabetic individuals, the SHAP value for FBG was mostly negative and gradually decreased across the range of 3.3–5.5 mmol/L. Beyond 5.5 mmol/L, the SHAP value increased and became positive around 6.3 mmol/L, suggesting a shift toward higher risk. In diabetic individuals, SHAP values for FBG were predominantly positive, increasing steadily within the 7–10 mmol/L range and stabilizing beyond 10 mmol/L, indicating a saturation point in its predictive contribution. Additional insights from Figure S4 also reveal a positive correlation between WtHR and age in both diabetic and non-diabetic populations.

At the individual level, force plots were employed to generate personalized predictions based on actual input feature values (Figure S5). These plots provide a simultaneous visualization of both the predicted probability of diabetes risk and the contribution of each feature to the model's output, facilitating intuitive and case-specific interpretation. In one case (Figure S5A), the model identified a diabetic patient as high-risk, accurately accounting for contextual clinical factors such as hypoglycemia caused by glucose-lowering medication, rather than relying solely on isolated biomarker values. In contrast, another case (Figure S5B) involves a healthy individual classified as low-risk, where the primary contributing factors were age and overweight status, despite a low overall predicted probability of diabetes.

Table 2. Results of risk stratification based on the development cohort

Dataset	Cohort (count)	High risk (proportion)	Moderate risk (proportion)	Low risk (proportion)	χ^2	padj
Training	diabetes (835)	476 (57.01%)	230 (27.54%)	129 (15.45%)	3352.90	<0.001
	healthy (6,530)	141 (2.16%)	628 (9.62%)	5761 (88.22%)		
Validation	diabetes (238)	126 (52.94%)	68 (28.57%)	44 (18.49%)	891.02	<0.001
	healthy (1,866)	39 (2.09%)	176 (9.43%)	1651 (88.48%)		
Testing	diabetes (119)	74 (62.19%)	37 (31.09%)	8 (6.72%)	540.96	<0.001
	healthy (933)	22 (2.36%)	92 (9.86%)	819 (87.78%)		

Webserver implementation

To enhance user accessibility to the DRPM, we developed the DRPM as a web-based tool, available at http://cbcb.cdutcm.edu.cn/drpm/. This web server allows users to input key health indicators, including FBG, age, WtHR, MSP, and BMI. Upon submission, the server calculates the individual's diabetes risk score and provides a corresponding risk alert. Additionally, the server generates a risk chart to visually display the individual's risk level. The chart uses a color-coded system: red for high risk, orange for moderate risk, and green for low risk. This intuitive interface offers users quick and easy access to their risk assessment, empowering them with useful information for proactive health management.

DISCUSSION

This study developed DRPM, a diabetes risk prediction model, aimed at screening for diabetes risk using NHANES data. Leveraging deep learning techniques, the proposed model classifies individuals into three distinct risk levels: high, moderate, and low, based on specific risk scores. These scores can be adapted for large-scale population screening during routine physical examination programs, as well as for personalized risk assessment and management. Moreover, the implementation of this DRPM extends beyond public health screening, offering potential integration with wearable devices and smart home IoT (internet of things) systems.[20] This would enable continuous monitoring of individual risk factors and facilitate early detection of diabetes.

The DRPM was constructed using five optimal features (FBG, age, WtHR, MSP, and BMI), which were identified through feature selection. Extensive research has shown a strong correlation between these features and the onset of diabetes. As widely recognized, FBG is a key marker for diagnosing diabetes, justifying its inclusion in the model. Previous studies have also highlighted the role of blood pressure management in diabetes prevention, although the exact mechanisms remain under investigation.[29] For instance, Qin et al. found a correlation between blood pressure levels and diabetes risk among hypertensive individuals, with lower systolic blood pressure levels linked to a reduced risk of developing diabetes.[30] Age is another significant risk factor for diabetes[31] demonstrating a positive association with diabetes incidence. BMI, a widely recognized screening tool, has long been considered an important factor in diabetes risk assessment.[32,33] Recent investigations have revealed that

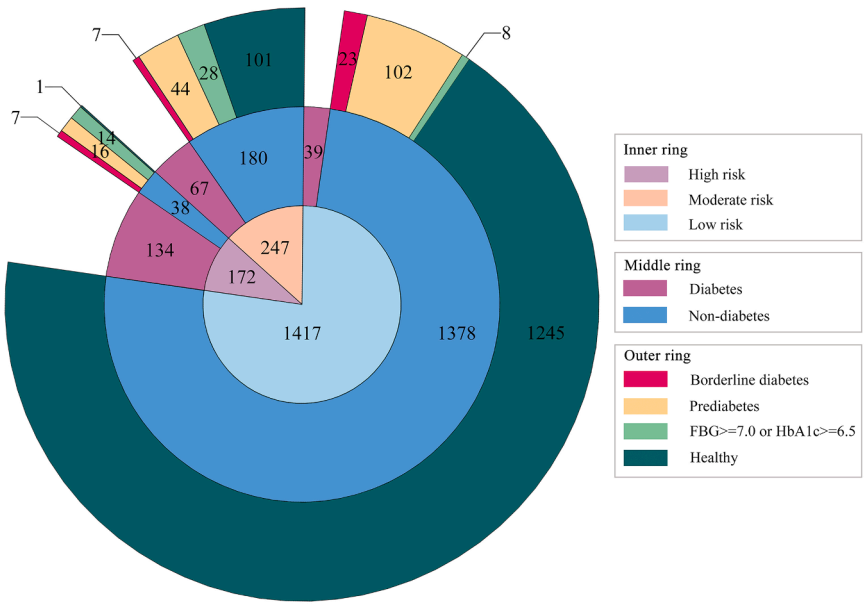


Figure 3. The risk groups among the external validation cohort predicted by the final model

The rings from the inner ring to the outer ring represent the following results: the innermost ring indicates the prediction result, the middle ring represents the actual classification of individuals, and the outermost ring shows the data analysis result.

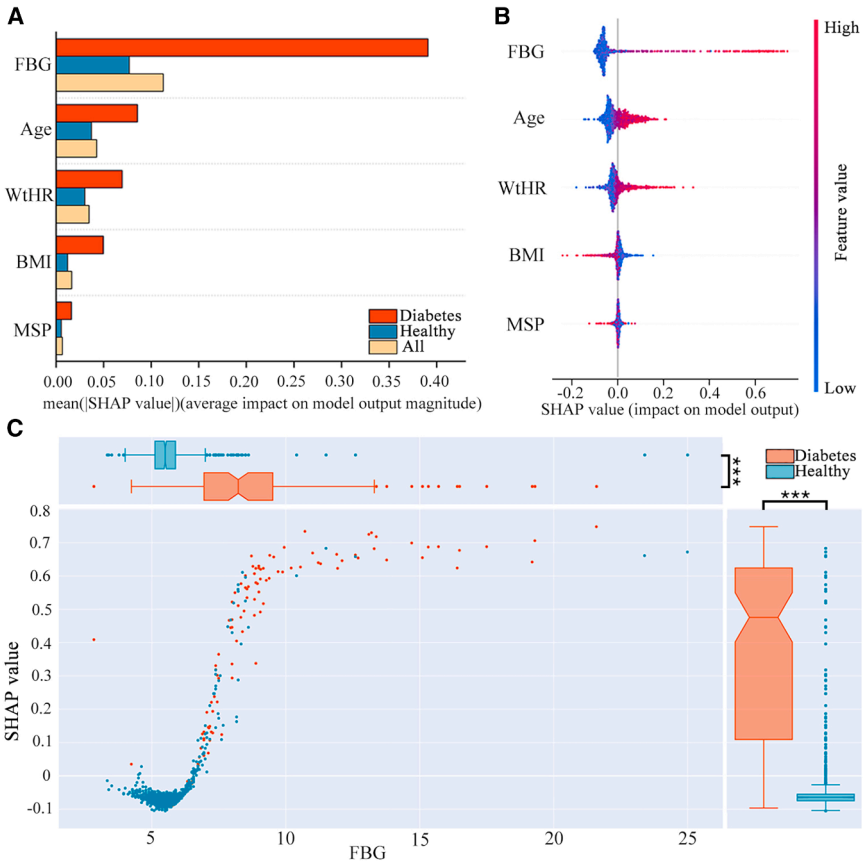


Figure 4. Global interpretation of the model

(A) Relative importance of the five important variables in the final model. The importance of each feature is evaluated based on its global contribution, and a similar ranking of importance is observed in both the diabetes and healthy datasets. (B) Global explainability of the model for the five features. This panel presents the global contribution of the top five most important features, ranked based on the mean of the absolute SHAP values. (C) Contextualized SHAP values as a function of variable values for FBG. Each dot represents an individual in the dataset. The color indicates whether the individual is classified as diabetic or healthy. The *x* axis corresponds to the real value of fasting blood glucose (FBG), while the *y* axis shows the SHAP value assigned to each individual for this specific feature. The SHAP value and FBG between diabetes and healthy were statistically significant ($p < 0.001$).

WtHR is also a critical factor in diabetes risk evaluation, with a 0.1 unit increase in WtHR associated with a 1.7-fold increase in diabetes risk.[34] These investigations suggest that the critical features identified in this model can serve as valuable indicators for diabetes risk.

Using the proposed model, the population was categorized into three risk groups based on risk indices of 0.5 and 0.13 for the segmented groups, respectively. Figures 2A–2C show that individuals with diabetes predominantly fall within the high-risk category. This stratification provides clinical guidance for individuals with varying risk profiles. Specifically, high-risk individuals can undergo timely examinations for early detection and treatment of diabetes, while those in the moderate-risk group can take proactive steps to modify their dietary and lifestyle habits to delay or prevent diabetes onset.

An independent external validation cohort was used to assess the robustness of the risk stratification model. As shown in Figure 3, the results from this cohort were consistent with the findings from the model development cohort. The majority of individuals with diabetes were correctly classified into the high-risk category. However, some non-diabetic individuals were assigned to both the high-risk and moderate-risk categories. Upon further investigation, it was found that many of these individuals had been informed by their physicians about their prediabetic status or had blood glucose or glycated hemoglobin levels already meeting the screening criteria for early-stage diabetes. These findings demonstrate the strong risk stratification ability of the model for non-diabetic individuals and confirm the model's potential role in supplementing diabetes screening efforts.

While the deep learning approach used in this study provides more detailed risk stratification for diabetes patients, several limitations must be acknowledged. First, the population data used were region-specific. Although external data were employed for validation, the validation process is not yet comprehensive. To ensure the model's accuracy, fairness, and robustness, further evaluation in diverse populations and settings is essential before its widespread clinical application.

In conclusion, we have developed a diabetes risk prediction model named DRPM using deep learning techniques, capable of analyzing extensive physical examination data to facilitate the early identification of individuals at risk for diabetes. The final model was built using five easily accessible features, simplifying data collection and eliminating the need for individuals to visit medical facilities for physical exams. This approach empowers individuals to engage in self-risk detection, contributing to enhanced public health efforts in diabetes screening and prevention. The model's user-friendly prerequisites make it a valuable tool for widespread diabetes risk assessment, with significant potential for improving early intervention and overall health outcomes.

MATERIALS AND METHODS

The model development cohort

The development cohort was obtained from the NHANES, covering data from 2011 to 2018 (Figure 5A). Diabetes status was determined

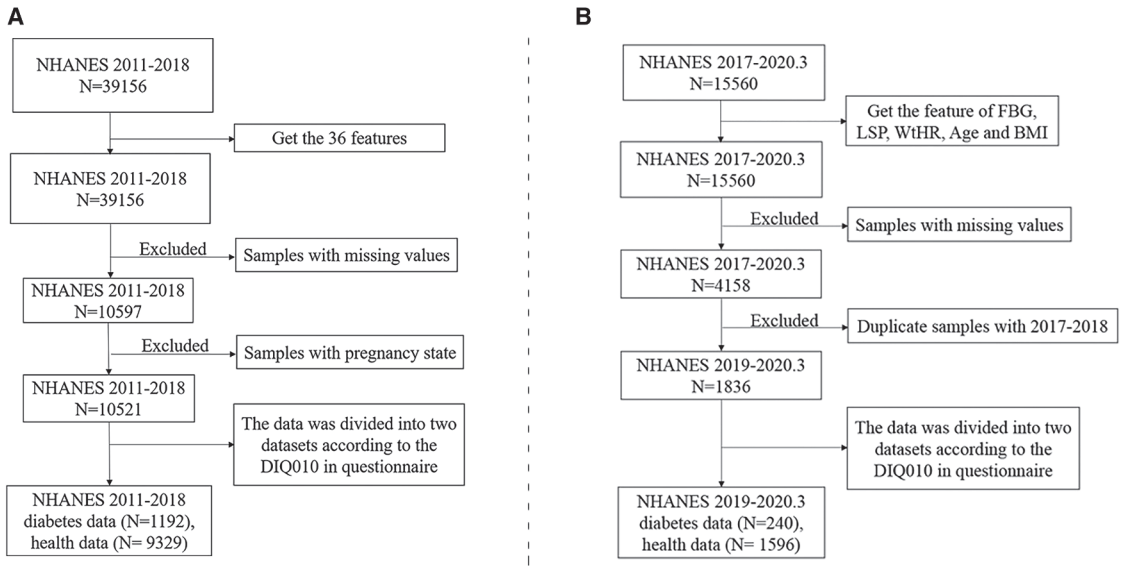


Figure 5. The flowchart illustrates the data processing steps. (A) The development cohort and (B) external validation cohort.

based on self-reported diagnosis, as indicated by a value of 1 in the DIQ010 (Doctor told you have diabetes, 1 means yes.) variable in the diabetes file about questionnaire data, which denotes that a doctor had diagnosed the individual with diabetes. Blood pressure readings were averaged from multiple measurements. To ensure data quality, records with missing values or those indicating pregnancy were excluded from the analysis. After data preprocessing, the final dataset consisted of 9,329 records from healthy individuals (53.52% male and 46.48% female) and 1,192 records from diabetic patients (48.86% male and 51.14% female). The age range for healthy individuals and diabetic patients was 12–80 years. Further details can be found in Table S4. The dataset was then randomly split into three subsets: the training set, validation set, and testing set, with a distribution ratio of 7:2:1.

The external validation cohort

The external validation cohort was sourced from the NHANES conducted between 2019 and March 2020 (Figure 5B). Diabetes in this cohort was defined based on self-reported diagnosis, indicated by a value of 1 in the DIQ010 variable within the questionnaire data. Blood pressure readings were averaged from multiple measurements. To ensure data integrity, samples with missing values were excluded, and duplicate samples from the 2017–2018 cohort were also removed. After data preprocessing, the external validation cohort comprised 240 records from individuals with diabetes and 1,596 records from individuals without diabetes, providing an independent validation for the developed model.

Feature extraction and fusion

A total of 33 primary features were extracted from the dataset, covering demographics, vital signs, and laboratory test data. In addition, three fusion features—WtHR, MSP, and mean diastolic pressure—were computed. Combining these primary and fusion features, a total of 36 features (listed in Table S5) were integrated into the preliminary set for developing the diabetes risk assessment system.

Feature selection

In deep learning models, it is common to aggregate a large number of features to provide the model with enough information for accurate predictions. However, in clinical settings, data collection is often constrained, and an excessive number of features can introduce redundancy or noise, potentially compromising the model's performance. Therefore, the key challenge is identifying the smallest subset of features that can yield optimal predictive results. To address this, we employed the feature significance ranking method for feature selection (Figure 6A). Initially, features were ranked based on their importance scores derived from the AUC and ANOVA.[35] The top ten features were selected for retention, and two inconsistent features (marked by the blue underline) were discarded (Figure S1). Redundant features were eliminated, and only those that were easily accessible and provided high relevance to the prediction were retained. This feature selection process helps mitigate the risks associated with high-dimensional overfitting while enhancing the practical applicability of the model. After screening, we identified the five optimal features from the initial set of 36.

Classification algorithms assessment

Four ML methods were compared: FCNN,[36] XGBoost,[37] LR,[38] and RF.[39] These models were selected for their distinct advantages in classification tasks. Each ML model was hyper-parameter tuned to ensure optimal performance and reliability (Table S6). The predictive performance of the four methods was evaluated based on the area under the ROC curve (AUC) for different feature combinations, with the goal of identifying the model with the best predictive capability (Figure 6B).

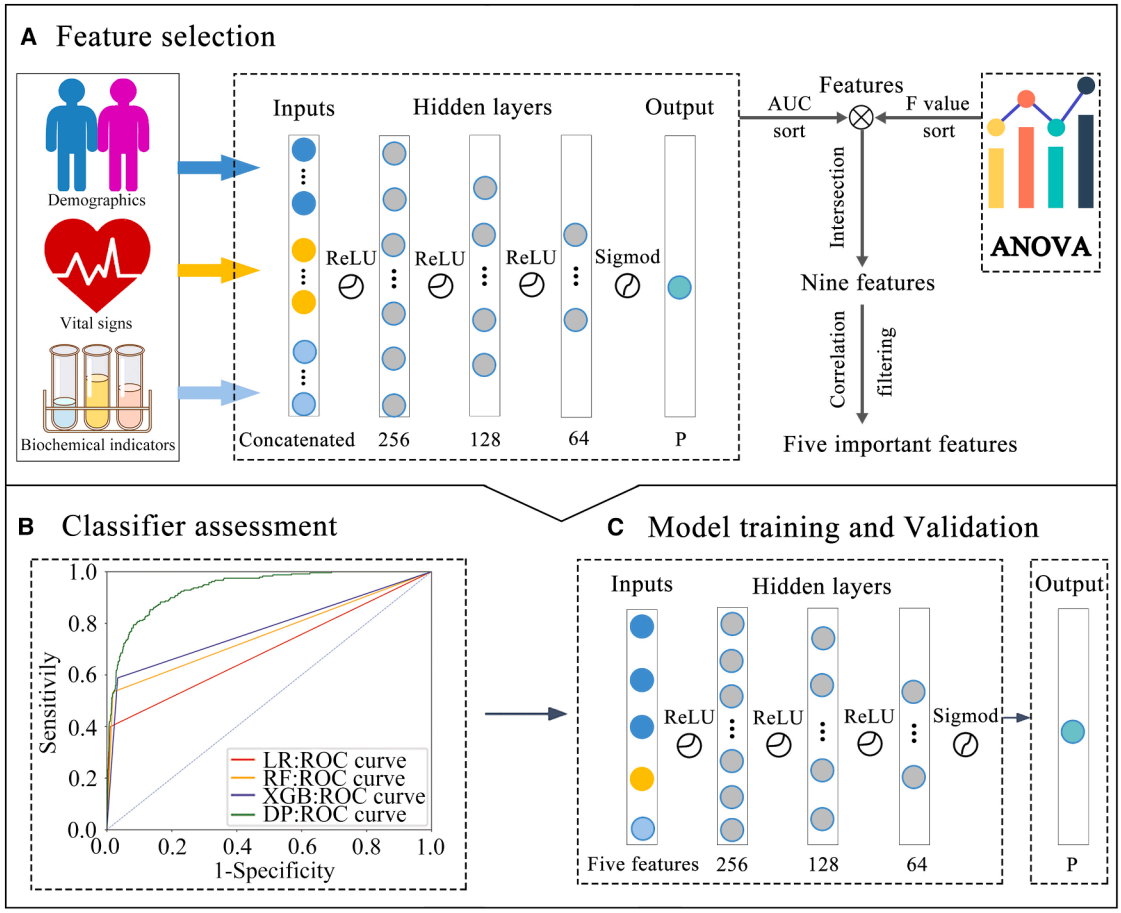


Figure 6. The diagram illustrates the analysis pipeline, highlighting the stages of data feature selection, classifier assessment, and model development

(A) The process of feature selection. (B) The process of classifier assessment. (C) The process of model development. In the output layer, PPP represents the magnitude of the diabetes risk probability: high risk when $p > 0.5$, medium risk when $0.13 < p \leq 0.5$, and low risk when $p \leq 0.13$.

Architecture for diabetes risk prediction model

For diabetes risk prediction, we developed a deep learning model using the PyTorch deep learning framework.[40] The model architecture consists of an input layer followed by three hidden layers with 256, 128, and 64 neurons, respectively (Figure 6C). The output layer is a linear 64-to-1 layer with a sigmoid activation function. The hidden layers use the rectified linear unit activation function[41] to introduce non-linearity, allowing the model to capture complex patterns in the data. The Adam optimization algorithm was employed as the optimizer for the model, with a learning rate of 1e−4. The batch size was set to the sample size, and the training procedure encompassed 800 epochs.

Model evaluation

The evaluation of model performance was carried out using accuracy, Mcc (Matthew's correlation coefficient), Sp (specificity), Sn (sensitivity, recall), precision, F1, and AUC. They were calculated as follows:

$$Accuracy = \frac{TP+TN}{TP+TN+FP+FN} \quad \text{(Equation 1)}$$

$$Mcc = \frac{TP \times TN - FP \times FN}{\sqrt{(TP+FP)(TP+FN)(TN+FP)(TN+FN)}} \quad \text{(Equation 2)}$$

$$Sp = 1 - FPR = \frac{TN}{TN+FP} \quad \text{(Equation 3)}$$

$$Sn = Recall = TPR = \frac{TP}{TP+FN} \quad \text{(Equation 4)}$$

$$\text{Precision} = \frac{TP}{TP+FP} \quad \text{(Equation 5)}$$

$$F1 = \frac{2 \times Precision \times Recall}{Precision+Recall} \quad \text{(Equation 6)}$$

In this context, TP (true positive) refers to the number of correctly predicted positive samples; FP (false positive) represents the number of negative samples incorrectly predicted as positive; FN (false negative) indicates the number of positive samples misclassified as

negative; and TN (true negative) represents the number of correctly predicted negative samples. AUC refers to the area under the ROC curve, with values ranging from 0 to 1. A higher AUC value signifies better model performance, with a value of 1 representing perfect discrimination.

DATA AVAILABILITY

The data used in this study are publicly available at NHANES (https://wwwn.cdc.gov/nchs/nhanes/).

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AUTHOR CONTRIBUTIONS

W.C. conceived and designed the study. F.N. conducted the experiments and implemented the algorithms. F.N. and X.S. performed the analysis and wrote the paper. All authors read and approved the final manuscript.

DECLARATION OF INTERESTS

The authors declare no competing interests.

SUPPLEMENTAL INFORMATION

Supplemental information can be found online at https://doi.org/10.1016/j.omtn.2025.102576.

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